

Exercise 1 **MPC Problem with Quadratic Costs**

1. There are two ways of formulating the problem:

- (i) The first approach includes the system dynamics as equality constraints with $x_1 - u_0 = 0.5x_0$ and $x_2 - 0.5x_1 - u_1 = 0$ and then solves the problem in the four variables $x = (x_1, x_2, u_0, u_1)$. The following MATLAB code implements this variant:

```
% options for quadprog
options = optimoptions('quadprog', 'Display', 'iter');

% initial conditions
x0 = 2;

% we optimize over [x1 x2 u0 u1]'
% cost function
H = eye(4);
f = zeros(4,1);

% matrices defining the inequality constraints
A = [-1  0  0  0;
      1  0  0  0;
      0  1  0  0;
      0 -1  0  0;
      0  0  1  0;
      0  0 -1  0;
      0  0  0 -1;
      0  0  0  1];

b = [-2.5; 5; 1; 1; 2; 2; 2; 2];

% matrices defining the equality constraints
Aeq = [ 1  0 -1  0;
       -0.5  1  0 -1];

beq = [0.5; 0] * x0;

% solving the problem via linprog
[x,FVAL,EXITFLAG,OUTPUT,LAMBDA] = quadprog(H, f, A, b, Aeq, beq, [], [], [], options);

% storing the solution
if( EXITFLAG > 0 )
    u0 = x(3);
    cost = FVAL + x0^2/2;
else
    u0 = NaN;
    cost = NaN;
end
```

Note that it is important to check whether quadprog found a feasible point by checking positivity of its output argument EXITFLAG.

- (ii) The second approach is to reduce the number of optimization variables by eliminating the states x_0, x_1 and x_2 in the problem formulation by

$$\begin{aligned}x_0 &= x(0), \\x_1 &= 0.5x(0) + u_0, \\x_2 &= 0.25x(0) + 0.5u_0 + u_1.\end{aligned}$$

The remaining optimization variables are u_0 and u_1 only:

$$\begin{aligned}f^*(x(0)) &= \min 0.5 \left(\frac{21}{16}x(0)^2 + (2.25u_0^2 + u_0u_1 + 2u_1^2) + (1.25x(0)u_0 + 0.5x(0)u_1) \right) \\ \text{s.t. } & 2.5 - 0.5x(0) \leq u_0 \leq 5 - 0.5x(0) \\ & -1 - 0.25x(0) \leq 0.5u_0 + u_1 \leq 1 - 0.25x(0) \\ & -2 \leq u_0 \leq 2 \\ & -2 \leq u_1 \leq 2\end{aligned}$$

Either way, the solution to the problem is given by

$$\begin{aligned}x_1^* &= 2.5, \\x_2^* &= 0.625, \\u_0^* &= 1.5, \\u_1^* &= -0.625, \\f^* &= 6.6406.\end{aligned}$$

2. For the first approach discussed above the KKT conditions can be written as

$$\begin{aligned}Hx^* + f + A^T\lambda^* + A_{eq}^T\nu^* &= 0 \\ Ax^* &\leq b, \quad A_{eq}x^* = b_{eq} \\ \lambda^* &\geq 0 \\ \lambda^{*T}(Ax^* - b) &= 0,\end{aligned}$$

whereby λ^* and ν^* are Lagrange Multipliers. These conditions are satisfied when plugging in the values obtained from quadprog's output LAMBDA:

$$\begin{aligned}\lambda^* &= \text{LAMBDA.ineqlin} = (4.3125, 0, 0, 0, 0, 0, 0, 0) \\ \nu^* &= \text{LAMBDA.eqlin} = (1.5, -0.625)\end{aligned}$$

Furthermore, one can deduce from λ^* that the only active constraint at the optimum is the constraint $x_1 \geq 2.5$.

3. The results for various initial states $x(0)$ are depicted in Figure 1. Infeasible solutions were assigned a value of NaN and are therefore not plotted. The set of admissible initial states (all $x(0)$ such that the problem is feasible) is $[1, 14]$.

It can be noted that the minimizer in the first time step, u_0^* , is piecewise affine (PWA) in $x(0)$. From the upper plot, we see that the cost function f^* is piecewise quadratic (PWQ) and convex in $x(0)$.

Note that for $1 \leq x(0) \leq 9$, we could express the optimal control law as an affine state feedback law, namely $u_0^*(x(0)) = -0.5x(0) + 2.5$. For the region $9 \leq x(0) \leq 14$, we get

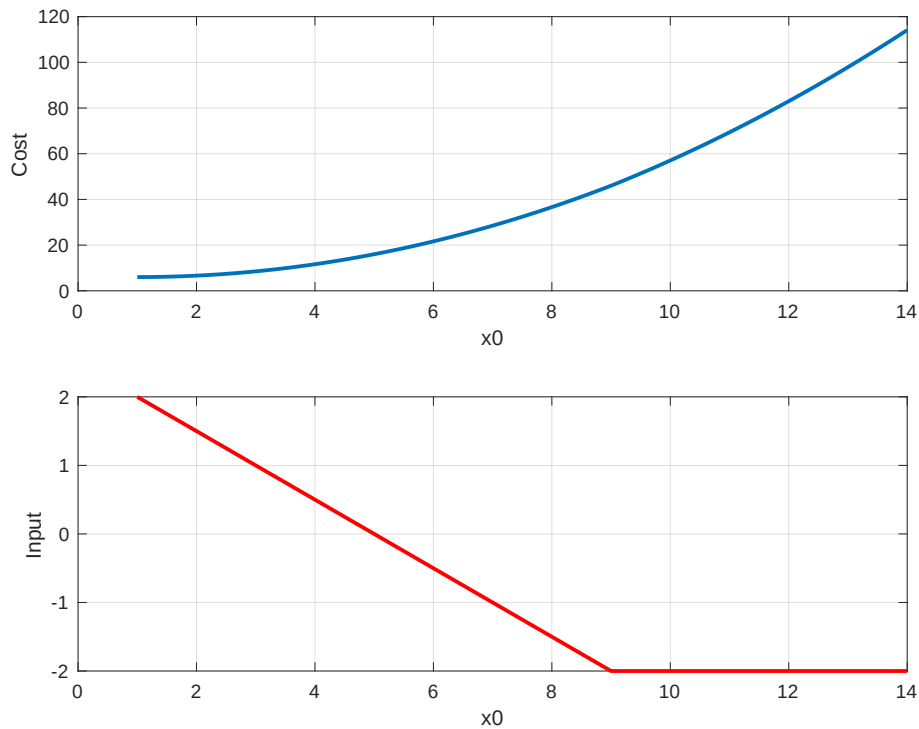


Figure 1: The optimal cost f^* (upper plot) and the first control input u_0^* (lower plot) as a function of the initial state $x(0)$.

$u_0^*(x(0)) = -2$. The change of control laws is due to different active constraints (try to verify this by calculating the Lagrange Multipliers!).

As we shall see later in the course, the idea behind *explicit MPC* is to exploit the PWA nature of $u_0^*(x(0))$ by precomputing the regions and the corresponding state feedback matrices offline, allowing for fast implementations of MPC.

4. Removing the lower bound on x_1 reduces the optimal cost value ($f^* = 2.2647$) since it allows for strong control action driving the states (x_1, x_2) towards zero. Removing the upper bound has no impact on the cost. Both conclusions can be derived from inspecting the Lagrange Multipliers of the corresponding constraints.
5. We would like to convert the Problem to an LP. See (Lec.4, Slide 35). First, we can express the equality constraints as

$$\begin{aligned} x_1 &= 0.5x_0 + u_0 \\ x_2 &= 0.5(0.5x_0 + u_0) + u_1 \end{aligned}$$

which means that the problem can be expressed in terms of u_0 and u_1 only. However, to get rid of the 1-norm, we can transform the problem by adding extra variables t_k and it becomes

$$\begin{aligned}
f^*(x(0)) &= |x(0)| + \min_{u_k, t_k} (t_1 + t_2 + t_3 + t_4) \\
\text{subject to } & -t_1 \leq x_1 \leq t_1 \\
& -t_2 \leq x_2 \leq t_2 \\
& -t_3 \leq u_0 \leq t_3 \\
& -t_4 \leq u_1 \leq t_4 \\
& 2.5 \leq x_1 \leq 5 \\
& -1 \leq x_2 \leq 1 \\
& -2 \leq u_0 \leq 2 \\
& -2 \leq u_1 \leq 2
\end{aligned}$$

We can replace x_1 and x_2 by their expressions and we have

$$\begin{aligned}
f^*(x(0)) &= \min_{u_k, t_k} (t_1 + t_2 + t_3 + t_4) \\
\text{subject to } & -t_1 \leq 0.5x_0 + u_0 \leq t_1 \\
& -t_2 \leq 0.25x_0 + 0.5u_0 + u_1 \leq t_2 \\
& -t_3 \leq u_0 \leq t_3 \\
& -t_4 \leq u_1 \leq t_4 \\
& 2.5 \leq 0.5x_0 + u_0 \leq 5 \\
& -1 \leq 0.25x_0 + 0.5u_0 + u_1 \leq 1 \\
& -2 \leq u_0 \leq 2 \\
& -2 \leq u_1 \leq 2
\end{aligned}$$

Setting the value $x(0) = 2$, we can write all the inequalities as a matrix inequality as

$$\begin{bmatrix}
-1 & 0 & 0 & 0 & 1 & 0 \\
-1 & 0 & 0 & 0 & -1 & 0 \\
0 & -1 & 0 & 0 & 0.5 & 1 \\
0 & -1 & 0 & 0 & -0.5 & -1 \\
0 & 0 & -1 & 0 & 1 & 0 \\
0 & 0 & -1 & 0 & -1 & 0 \\
0 & 0 & 0 & -1 & 0 & 1 \\
0 & 0 & 0 & -1 & 0 & -1 \\
0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 0.5 & 1 \\
0 & 0 & 0 & 0 & -0.5 & -1 \\
0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & -1
\end{bmatrix}
\begin{bmatrix}
t_1 \\
t_2 \\
t_3 \\
t_4 \\
u_0 \\
u_1
\end{bmatrix}
\leq
\begin{bmatrix}
-1 \\
1 \\
-0.5 \\
0.5 \\
0 \\
0 \\
0 \\
0 \\
4 \\
-1.5 \\
0.5 \\
1.5 \\
2 \\
2 \\
2 \\
2
\end{bmatrix}$$

The following MATLAB code implements the solution:

```

clear all
close all
clc

% options for linprog
options = optimoptions('linprog', 'Display', 'iter');

% initial conditions
x0 = 2;

% we optimize over [eps1 eps2 eps3 eps4 u0 u1]'
% cost function
f = [1 1 1 1 0 0]';

% inequality constraint
Geps = [-1 0 0 0;...
        -1 0 0 0;...
         0 -1 0 0;...
         0 -1 0 0;...
         0 0 -1 0;...
         0 0 -1 0;...
         0 0 0 -1;
         0 0 0 -1];

Gu = [1 0;...
      -1 0;...
      0.5 1;...
      -0.5 -1;...
      1 0;...
      -1 0;...
      0 1;...
      0 -1];

G = [1 0;...
     -1 0;...
     0.5 1;...
     -0.5 -1;...
     1 0;...
     -1 0;...
     0 1;...
     0 -1];

Eeps = [-0.5; 0.5; -0.25; 0.25; 0; 0; 0; 0];
Eu = [-0.5; +0.5; -0.25; 0.25; zeros(4,1)];
E = [Eeps; Eu];
w = [zeros(8,1); 5; -2.5; 1; 1; 2; 2; 2];

% matrix that we will be feeded to linprog
A = [Geps Gu; zeros(8,4) G];
b = E*x0 + w;

% solving the problem via linprog
[x,FVAL,EXITFLAG,OUTPUT,LAMBDA] = linprog(f, A, b, [], [], [], [], [], options);

% storing the solution
if( EXITFLAG > 0 )
    u0 = x(5) ;
    cost = FVAL + abs(x0);
else
    u0 = NaN;
    cost = NaN;
end

```

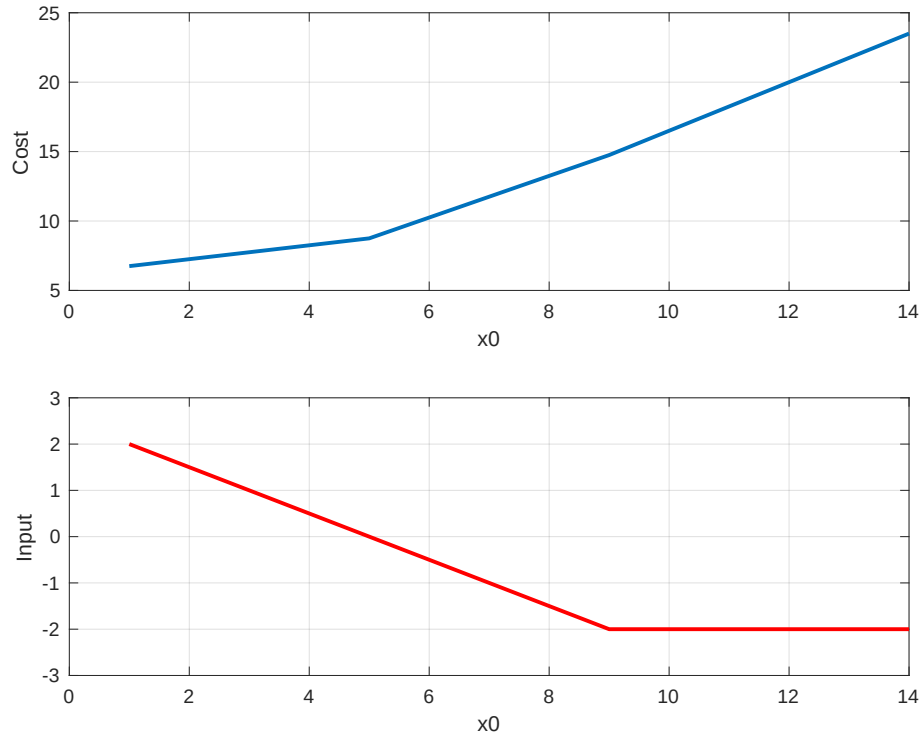


Figure 2: The optimal cost f^* (upper plot) and the first control input u_0^* (lower plot) as a function of the initial state $x(0)$.

The solution to the problem is given by

$$\begin{aligned}
 x_1^* &= 2.5, \\
 x_2^* &= 0, \\
 u_0^* &= 1.5, \\
 u_1^* &= -1.25, \\
 f^* &= 7.25.
 \end{aligned}$$

6. The results for various initial states $x(0)$ are depicted in Figure 2. Infeasible solutions were assigned a value of NaN and are therefore not plotted. The set of admissible initial states (all $x(0)$ such that the problem is feasible) is $[1, 14]$.

It can be noted that the minimizer in the first time step, u_0^* , is piecewise affine (PWA) in $x(0)$. From the upper plot, we see that also the cost function f^* is piecewise affine (PWA) and convex in $x(0)$.

7. Removing the lower bound on x_1 reduces the optimal cost value ($f^* = 3$) since it allows for strong control action driving the states (x_1, x_2) towards zero. As before, removing the upper bound has no impact on the cost.

Exercise 2 1-Norm, ∞ -Norm

- $p = 1$: By definition of the 1-norm we get

$$\min \|Ax\|_1 = \min \sum_{i=1}^N |a_i^T x| = \min_{z_i = |a_i^T x|, i=1, \dots, N} \sum_{i=1}^N z_i.$$

The latter problem has decision variables x and z and is non-convex, so not an LP yet! The crucial observation now is that the (non-convex) equality constraints $z_i = |a_i^T x|$, $i = 1, \dots, N$, can be relaxed to $|a_i^T x| \leq z_i$, $i = 1, \dots, N$ without changing the minimum/minimizer. To see this, we construct the following contraction:

Assume that at the minimum an inequality constraint is *not* active, i.e. $\exists k : |a_k^T x| < z_k$. In this case we could get a lower value than the minimum which contradicts the assumption that we are at the minimum. We conclude that although we solve the relaxed problem, at the minimum the equalities are recovered, hence

$$\min \|Ax\|_1 = \min_{|a_i^T x| \leq z_i, i=1, \dots, N} \sum_{i=1}^N z_i = \min_{-z_i \leq a_i^T x \leq z_i, i=1, \dots, N} \sum_{i=1}^N z_i ,$$

which leads to the LP in the required format (using the definition $y := (x, z)$)

$$\begin{aligned} & \min [\mathbf{0}_{1 \times n}, \mathbf{1}_{1 \times N}] y \\ & \text{s.t.} \quad \begin{bmatrix} A & -I \\ -A & -I \end{bmatrix} y \leq \mathbf{0} . \end{aligned}$$

- $p = \infty$: We take a different approach here, called the *epigraph reformulation* of the problem:

$$\min \|Ax\|_\infty = \min_{\|Ax\|_\infty \leq t} t .$$

By the definition of the ∞ -norm we can rewrite the constraint as

$$\|Ax\|_\infty \leq t \iff \max_{i=1, \dots, N} |a_i^T x| \leq t \iff -t \leq a_i^T x \leq t, \quad i = 1, \dots, N ,$$

such that we arrive at the LP (using the definition $y := (x, t)$)

$$\begin{aligned} & \min [\mathbf{0}_{1 \times n}, 1] y \\ & \text{s.t.} \quad \begin{bmatrix} A & -1 \\ -A & -1 \end{bmatrix} y \leq \mathbf{0} . \end{aligned}$$

Note: The 1-norm reformulation requires more auxiliary variables than the ∞ -norm reformulation.